

DRAFT

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Question #: 1

A 24-year-old man immunocompromised man is being treated for a chronic abscess in the ischioanal fossa. On further investigation, another large abscess was identified located on the anterior surface of the psoas major. On physical examination, there was loss of the cremasteric reflex. Which of the following areas will have reduced sensation due to the location of this abscess?

- A. Anterior aspect of the scrotum
- B. Skin over the femoral triangle
- C. Antero-lateral thigh
- D. Hypogastric region of the abdominal wall
- E. Superior aspect of the gluteal region

Question #: 2

Following an emergency appendectomy a patient who complained of having numbness (paraesthesia) of the skin in the suprapubic region of the abdomen. The most likely nerve(s) that has been injured during the operation is:

- A. Genitofemoral
- B. Iliohypogastric
- C. Subcostal
- D. Lateral femoral cutaneous and iliohypogastric
- E. Ilioinguinal and iliohypogastric
- F. Ilioinguinal

Question #: 3

A 30-year-old man comes to the ER with excruciating abdominal pain. The on-call surgeon decides to perform an exploratory laparotomy. Where would the incision most likely be made to separate the left and right rectus sheaths?

- A. Midaxillary line
- B. Linea alba
- C. Semilunar line
- D. Arcuate line
- E. Tendinous intersection

Question #: 4

A 55-year-old man comes to the clinic with a bulge and discomfort in his groin. A diagnosis of indirect inguinal hernia is made, and he is admitted to the hospital. During an open hernioplasty, the spermatic cord is identified. Which of the following nerves provides innervation to component of the spermatic cord is derived from the internal abdominal oblique muscle?

- A. Femoral branch of the genitofemoral
- B. Ilioinguinal
- C. Iliohypogastric
- D. Genital branch of genitofemoral
- E. Femoral

Question #: 5

A group of four 3rd year medical students are currently rotating in surgery. The general surgeon reviews their knowledge of the abdominal wall before having them observe a cholecystectomy. During a discussion on the blood supply, the general surgeon asks ; "Which vessel most likely gives rise to the inferior epigastric artery?"

- A. External iliac
 - B. Internal iliac
 - C. Femoral
 - D. Abdominal aorta
 - E. Deep circumflex iliac
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Question #: 6

A 25-year-old woman comes to the physician because of a 5-month history of pain and tingling in the upper thigh. Physical examination shows decreased sensation in the medial aspect of her thigh and difficulty adducting her thigh. Imaging studies show an enlarged ovary, compressing a nerve on the lateral wall of the pelvis. Which of the following muscles is most likely innervated by this nerve?

- A. Obturator internus
- B. Obturator externus
- C. Gluteus maximus
- D. Piriformis
- E. Levator ani

Question #: 7

A 59-year-old man comes to the physician because of a 4-hour history of sharp pain and vesicular lesions over his left thoracic wall extending to the left nipple line. He has a history of chickenpox infection in childhood. Which of the following structures most likely contains the latent virus that has reactivated and is responsible for this patient's condition?

- A. T7 dorsal root ganglion
- B. T10 dorsal root ganglion
- C. T7 ventral root
- D. T4 ventral root
- E. T4 dorsal root ganglion

Question #: 8

A 34-year-old woman comes to the physician for a routine pelvic examination. The physician enquires about the muscles that exit the pelvis through the greater sciatic foramen from a group of medical students. Which of the following muscles is most likely being asked for by the physician?

- A. Piriformis

- B. Coccygeus
- C. Pubococcygeus
- D. Obturator internus
- E. Psoas major

Question #: 9

A 45-year-old man comes to the emergency department because of 2-week history of groin pain and a palpable mass just superior to the inguinal ligament. He is diagnosed with a hernia and is prepped for surgery. During the operation, intestines are found in the deep inguinal ring. Which of the following is most likely the hernia found in this patient?

- A. Direct inguinal
- B. Umbilical
- C. Femoral
- D. Lumbar
- E. Indirect inguinal

Question #: 10

A 55-year-old-man is brought to the emergency department by paramedics after he was involved in a motor vehicle collision. His blood pressure is 100/65mmHg, pulse is 92/min and respirations are 17/min. Physical examination shows pink and moist mucous membranes. He has bruising and tenderness to the left upper quadrant and suprapubic region of his abdomen. During surgery a paramedian incision is attempted with medial retraction of the rectus abdominis muscle. This results in the rupture of the inferior epigastric artery. A branch of which of the following vessels will most likely secure collateral circulation to the muscle?

- A. Internal thoracic
 - B. External iliac
 - C. Superficial epigastric
 - D. Periumbilical
 - E. Intercostal
-

Question #: 11

A 1-month-old boy is brought to the physician by his mother because of a 2-week history of a swelling in his right scrotum. Transillumination shows no masses but a fluid-filled space. A diagnosis of hydrocele is made. The accumulation of fluid is in the tunica vaginalis. Which of the following abdominal layers most likely gives rise to this structure?

- A. External oblique muscles
- B. Internal obliques aponeurosis
- C. Transversus abdominus muscle
- D. Transversalis fascia
- E. Parietal peritoneum

Question #: 12

A 61-year-old man comes to the physician because of a 2-month history of weakness in his right knee. Physical examination shows weakness of knee extension. Palpation of the abdominal wall in the lower right quadrant shows a firm mass verified by imaging to be located in the right iliac fossa on the lateral border of the psoas major muscle. Given the location and the symptoms, which of the following nerves in the lumbar plexus is most likely involved?

- A. Obturator
- B. Lumbosacral trunk
- C. Iliohypogastric
- D. Ilioinguinal
- E. Femoral

Question #: 13

A 45-year-old woman comes to the physician because of a 1-week history of bowel incontinence. Injury to which of the following muscles is most likely responsible for her condition?

- A. Pubococcygeus
- B. Ilicoccygeus
- C. Coccygeus
- D. Obturator internus

Question #: 14

A 35-year-old woman comes to the physician because of a 1-week history of a lump in her left breast. Physical examination shows a 1-cm mass in the upper outer quadrant of her left breast. Examination of a biopsy specimen of the mass shows cancerous changes. To which of the following groups of lymph nodes would the cancer most likely metastasize?

- A. Anterior axillary (pectoral)
- B. Lateral axillary (humeral)
- C. Parasternal
- D. Central axillary
- E. Apical (infraclavicular)

Question #: 15

A 25-year-old man is brought to the emergency department by his friends because of a 3-hour history of severe pain in the perineum. Physical examination shows a tender, erythematous swelling near the anal opening. A diagnosis of an anal abscess is made. During surgical debridement of the abscess, the physician is careful not to damage branches of the peripheral nerve that supply the skin and muscles of the perineum. Which of the following nerve fibers are most likely found in this nerve?

- A. Somatic afferents, visceral afferents and preganglionic parasympathetics
- B. Somatic efferents, somatic afferents and preganglionic sympathetics
- C. Preganglionic sympathetics, postganglionic sympathetics and somatic afferents
- D. Postganglionic sympathetics, somatic afferents and somatic efferents
- E. Postganglionic parasympathetics, somatic afferents and somatic efferents

Question #: 16

A 21-year-old woman who is a university student is suspected of having meningitis. Since this condition may alter the chemical composition of cerebrospinal fluid, a lumbar puncture is performed. Between which two vertebrae is the most appropriate site to insert the needle in this patient?

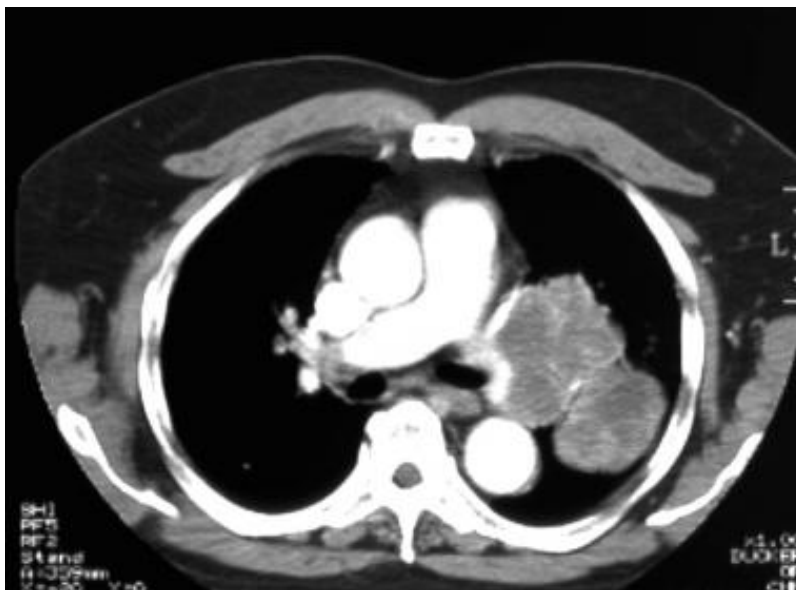
- A. T11-T12
- B. T12-L1
- C. L1-L2
- D. L2-L3
- E. L3-L4

Question #: 17

A 50-year-old man is brought to the physician because of a 6-month history of cough, chest pain and unintentional weight loss. The physician orders several imaging studies, one of which is shown. This is an example of which of the following imaging modalities?

- A. Radiography
- B. Magnetic resonance imaging
- C. Computer tomography
- D. Ultrasonography
- E. Angiography

Attachment:



Lecture 77.jpg

Question #: 18

A 54-year-old man comes to the physician because of 3-month history of weakness in his fingers. Physical examination shows difficulty flexing the distal interphalangeal joints of the index and middle fingers. Pronation and wrist flexion are weakened. There are no sensory deficits. Which of the following nerves is most likely affected?

- A. Ulnar nerve at Guyon's canal
- B. Anterior interosseous nerve between the two heads of pronator teres
- C. Median nerve in the carpal tunnel
- D. Median nerve at the supracondylar ridge
- E. Posterior interosseous nerve beneath the supinator

Question #: 19

A 78-year old woman comes to the physician because of a 1-month history of worsening stiffness of her fingers. During the physical examination, the physician holds the four fingers of her hand together and asks her to spread her fingers. Which of the following muscles is most likely being tested??

- A. Lumbricals
- B. Palmar interossei
- C. Dorsal interossei
- D. Flexor pollicis brevis
- E. Abductor pollicis brevis

Question #: 20

A 32-year-old woman comes to the physician because of a 1-day history of pain and swelling of her pinky finger. Three days ago, she sustained a cut to her finger while chopping vegetables. Physical examination shows an erythematous pinky finger. Which of the following groups of axillary lymph nodes will most likely first receive drainage from the site of the infection?

- A. Apical
- B. Anterior (pectoral)
- C. Posterior (scapular)
- D. Lateral (humeral)

Question #: 21

A 15-year-old girl is brought to the emergency department by her parents because of a 3-hour history of falling on her outstretched hand. Physical examination shows tenderness on palpation of the anatomical snuff box. An x-ray of her hand shows no obvious fractures however, the physician treats the patient by splinting the area. Which of the following bones is most likely fractured?

- A. Lunate
 - B. Scaphoid
 - C. Hamate
 - D. 1st metacarpal
 - E. Triquetrum
-

Question #: 22

A 36-year-old woman comes to the physician because of a 2-month history of difficulty moving her upper limb. MRI of the shoulder shows a tear in the supraspinatus muscle. Which of the following movements of the shoulder joint is most likely affected?

- A. Flexion
 - B. Adduction
 - C. Lateral rotation
 - D. Abduction
 - E. Medial rotation
-

Question #: 23

A 25-year-old man comes to the physician because of a 2-month history of pain in his elbow while attempting to swing his golf club at his weekly game. Physical examination shows tenderness over the medial epicondyle. Which of the following structures is most likely inflamed?

- A. Common extensor tendon

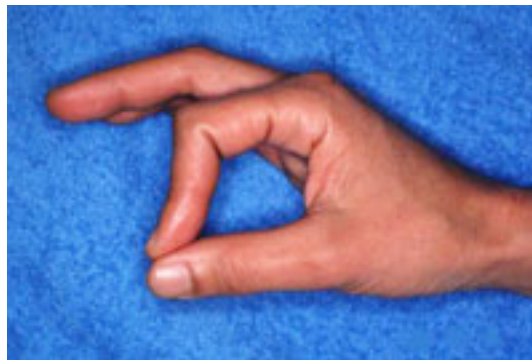
- B. Transverse carpal ligament
- C. Palmar aponeurosis
- D. Common flexor tendon
- E. Bicipital aponeurosis

Question #: 24

A 16-year-old girl is brought to the physician by her mother because of a 2-month history of difficulty holding her pen between her thumb and index finger. Physical examination shows no sensory deficit and when asked to make the "OK" sign, her fingers are positioned as shown in the attached image. Her symptoms are most likely caused by a loss of innervation to which of the following pairs of muscles?

- A. Flexor digitorum superficialis and flexor pollicis brevis
- B. Extensor pollicis longus and extensor pollicis brevis
- C. Dorsal interossei and palmar interossei
- D. Adductor pollicis and abductor pollicis longus
- E. Flexor digitorum profundus and flexor pollicis longus

Attachment:



anterior_interosseous_nerve_palsy_01.jpg

Question #: 25

A 27-year-old man is brought to the emergency department because of an injured left arm after losing control of his motorcycle. Physical examination shows an inability to abduct his arm to the horizontal, and a diagnosis of axillary nerve injury is made. This injury will most likely result in sensory findings in which of the following areas?

- A. Upper lateral aspect of the arm
- B. Upper medial aspect of the arm
- C. Lateral aspect of the forearm
- D. Medial aspect of the forearm
- E. Posterior aspect of the arm

Question #: 26

A 68-year-old man comes to the physician because of a history of back pain that irradiates to his right thigh. He is diagnosed with a posterior abdominal wall abscess which affects the iliopsoas muscle. Which of the following functions will most likely be affected in this patient?

- A. Medial rotation of the thigh
- B. Extension of the thigh
- C. Flexion at the knee
- D. Lateral rotation of the thigh
- E. Flexion of the thigh

Question #: 27

A 40-year-old man comes to the physician for a follow-up examination because of ambulation difficulty following a motor vehicle collision. Physical examination shows he falls towards his right side when attempting to stand on his left foot. Which of the following nerves has most likely been injured?

- A. Left obturator
- B. Left inferior gluteal nerve
- C. Right inferior gluteal nerve
- D. Left superior gluteal nerve
- E. Right superior gluteal nerve

Question #: 28

A 36-year-old man comes to the emergency department because of a 2-day history of pain and tenderness in his right wrist after a fall on his outstretched hand. Physical examination shows

tenderness on the lateral side of the wrist, just proximal to the base of the first metacarpal. Which of the following is the most likely diagnosis?

- A. Fracture of the fifth metacarpal
- B. Fracture of the hamate
- C. Tenosynovitis of thumb extensors
- D. Fracture of the scaphoid
- E. Tenosynovitis of the long flexor tendons

Question #: 29

A 60-year-old multiparous woman comes to the physician because of a 6-month history of episodes of fecal incontinence. This is due to weakness in one of the muscles of the pelvis. A tear in which of the following muscles would most likely result in this patient's symptoms?

- A. Iliococcygeus
- B. Piriformis
- C. Obturator internus
- D. Coccygeus
- E. Puborectalis

Question #: 30

A 45-year-old man comes to the clinic because of a 2-week history of a bulge in his right groin. He reports that the bulge increases in size when standing or coughing but reduces when lying down. His temperature is 36.8°C (98.2°F), pulse is 76/min, respirations are 18/min, and blood pressure is 128/82 mm Hg. Physical examination shows a soft, non-tender, reducible mass in the right inguinal region that is lateral to the inferior epigastric vessels. Which of the following is the most likely diagnosis?

- A. Direct inguinal hernia
- B. Indirect inguinal hernia
- C. Femoral hernia
- D. Spigelian hernia
- E. Obturator hernia